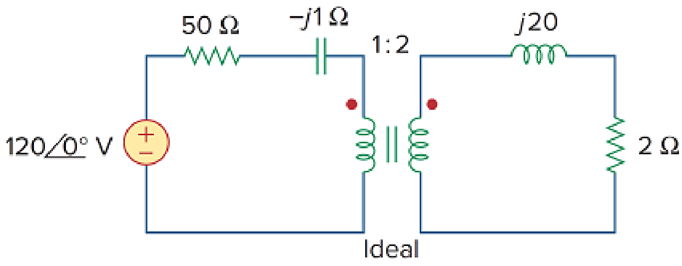
Chapter 13, Problem 42P

Problem

For the circuit in Fig. 13.107, determine the power absorbed by the 2-Ω resistor. Assume the 120 V is an rms value.

Figure 13.107



Step-by-step solution

Step 1 of 6

Refer to Figure 13.107 in the textbook for the actual circuit.

Redraw the circuit as follows.

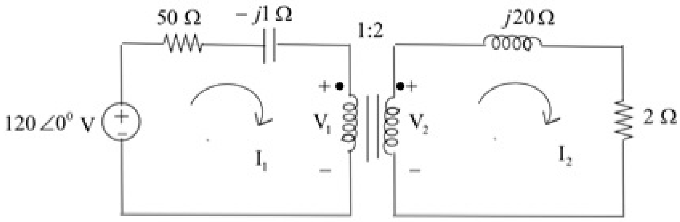


Figure 1

Step 2 of 6

Apply Kirchhoff's voltage law to mesh 1.

$$(50 - j1)I_1 + V_1 - 120 = 0$$

$$(50 - j1)I_1 + V_1 = 120 \quad \text{......} \ (1)$$

Apply Kirchhoff's voltage law to mesh 2.

$$(2 + j20)I_2 - V_2 = 0 \quad \text{......} \ (2)$$

Step 3 of 6

At the transformer terminals,

$$\frac{V_2}{V_1}=\frac{N_2}{N_1}$$

Substitute 2 for N_2, 1 for N_1.

$$\frac{V_2}{V_1}=\frac{2}{1}$$
$$V_1=\frac{V_2}{2}$$

The primary and secondary currents are related to the turns ratio in the inverse manner.

$$\frac{I_1}{I_2}=\frac{N_2}{N_1}$$

Substitute 2 for N_2, 1 for N_1.

$$\frac{I_1}{I_2}=\frac{2}{1}$$
$$I_1=2I_2$$

Step 4 of 6

Substitute $\frac{V_2}{2}$ for V_1 and $2I_2$ for I_1 in equation (1).

$$(50-j1)2I_2+\frac{V_2}{2}=120 \quad \ldots\ldots (3)$$

Multiply equation (3) by 2,

$$(200-j4)I_2+V_2=240 \quad \ldots\ldots (4)$$

Step 5 of 6

Add equations (2) and (4).

$$(2+j20)I_2 \ -V_2+(200-j4)I_2+V_2=240$$
$$(202+j16)I_2 \ =240$$
$$I_2=\frac{240}{202+j16}$$
$$I_2=1.184\angle-4.53^\circ \text{ A}$$

Step 6 of 6

Calculate the power absorbed by the 2Ω resistor.

$$P=|I_2|^2 R$$

Substitute 1.184 A for $|I_2|$, 2Ω for R.

$$\begin{aligned}P&=(1.184)^2(2)\\&=(1.4028)(2)\\&=2.805\text{ W}\end{aligned}$$

Therefore, the power absorbed by the 2Ω resistor is $\boxed{2.805\text{ W}}$.